AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 12 · STUDENT LESSON

Accuracy as a Fraction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Count the matches. Name the denominator. Inspect the errors.

Compare overall and class-specific scores without hiding rare-class failures or missing coverage.

Keep one habit: Simple accuracy → labelled matrix → class denominators → tradeoff and evidence limit.

MAIN LESSON · SECTION 1

What does an accuracy percentage leave out?

> Start with correct divided by evaluated, then inspect the two classes.

Your learning goals

I can explain why overall **accuracy** can hide failure on a less common class.

I can compare a **training-majority baseline** with class-specific results.

I can calculate an equal-class average and report its limits alongside the confusion matrix.

Warm up with a simple denominator

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- If 3 of 12 responses are Yes, what fraction are Yes?

Your end goal

By the end, I can compare **accuracy** with class-specific performance in an imbalanced evaluation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

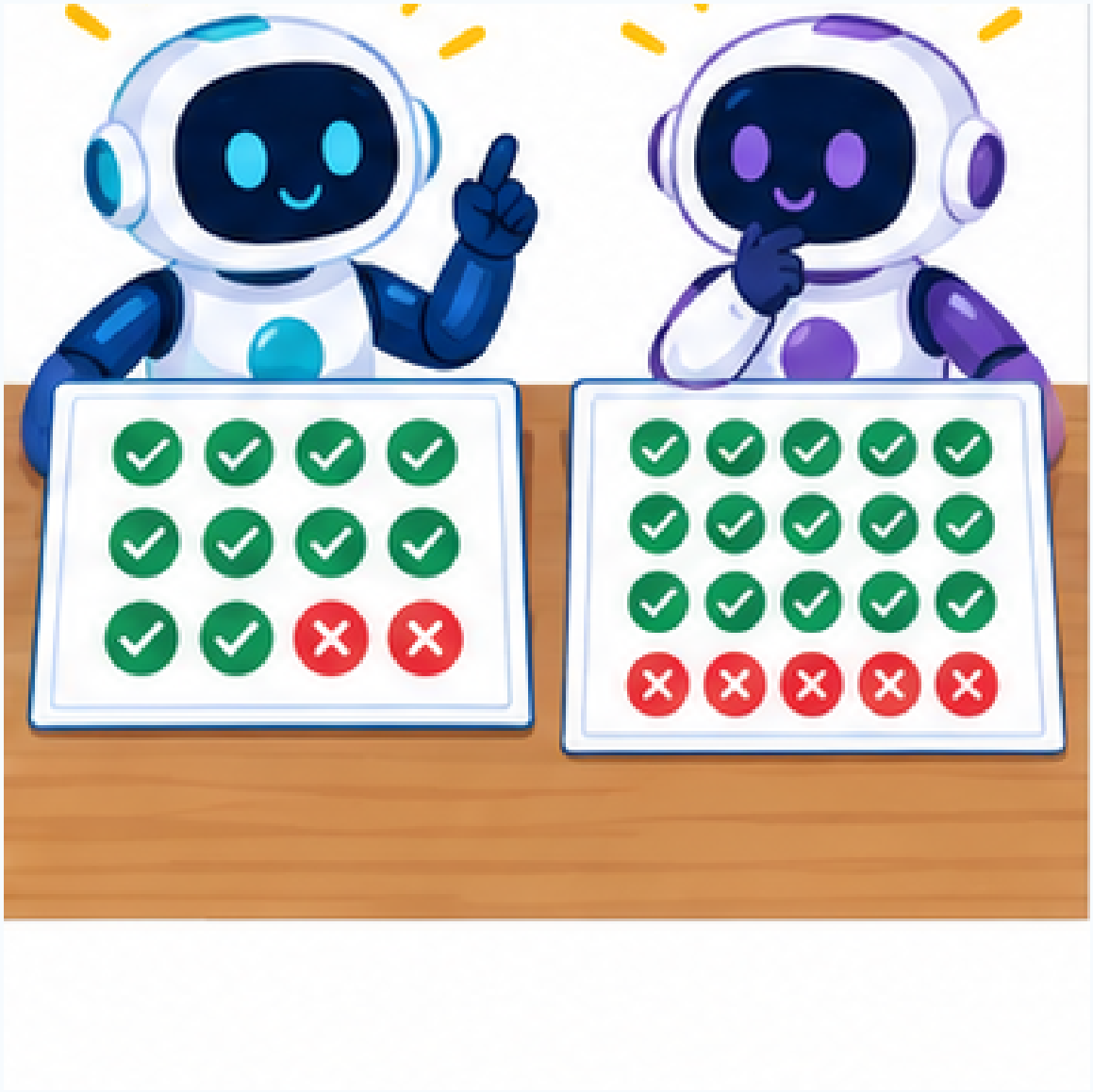
Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 12: **Accuracy** as a Fraction

Accuracy is correct **prediction**s divided by all **prediction**s.
</details>



THINKER MISSION

Calculate and compare 7/10 and 18/20 models.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

With reference rows and **prediction** columns ordered clear then blocked, where are false negatives when blocked is positive?

Readiness for Summaries with counts and proportions

> **My readiness explanation**

This week’s end goal

MAIN LESSON · SECTION 2

Count correct cases, then read a labelled matrix

> Keep the positive class and axis order visible.

First calculate one accuracy fraction

Start small: 9 correct predictions out of 12 evaluated cases gives accuracy 9/12=3/4=75%. The denominator includes the three errors. Now ask which examples were wrong. Two classifiers can have the same total correct while failing on different kinds of cases.

Learn the four matrix cells

Read the matrix like a labelled counting grid. Each row names the reference class; each column names the predicted class. Here both axes are clear, then blocked. Clear→clear and blocked→blocked are correct. Clear→blocked is a false positive (a false alarm); blocked→clear is a false negative (a missed blocked case). Positive means the class we choose to track, not good or desirable. We introduce these names here; no future lesson is a prerequisite.

Why class imbalance matters

Accuracy counts all correct **prediction**s and divides by all evaluated examples. Each example has equal weight, so a common class contributes more to the score. A high result can hide failure on a rare class.

This week keeps the confusion matrix visible and checks both classes separately.

The training-derived baseline and evaluation table

New fictional **training** data has 16 clear and 4 blocked labels. Its **majority baseline** predicts clear for every query; choose that rule from **training**, not evaluation labels.

A separate evaluation set also contains 16 clear and 4 blocked cases. A frozen candidate and the baseline are evaluated on the same 20 records.

Method and reference row	Predicted clear	Predicted blocked
Baseline reference clear	16	0
Baseline reference blocked	4	0
Candidate reference clear	12	4
Candidate reference blocked	0	4

Explain the equal-score tradeoff

Rows are references and columns are **prediction**s, ordered clear then blocked. Blocked is positive. Both methods get 16/20=80 percent **accuracy**. The baseline misses all four blocked cases.

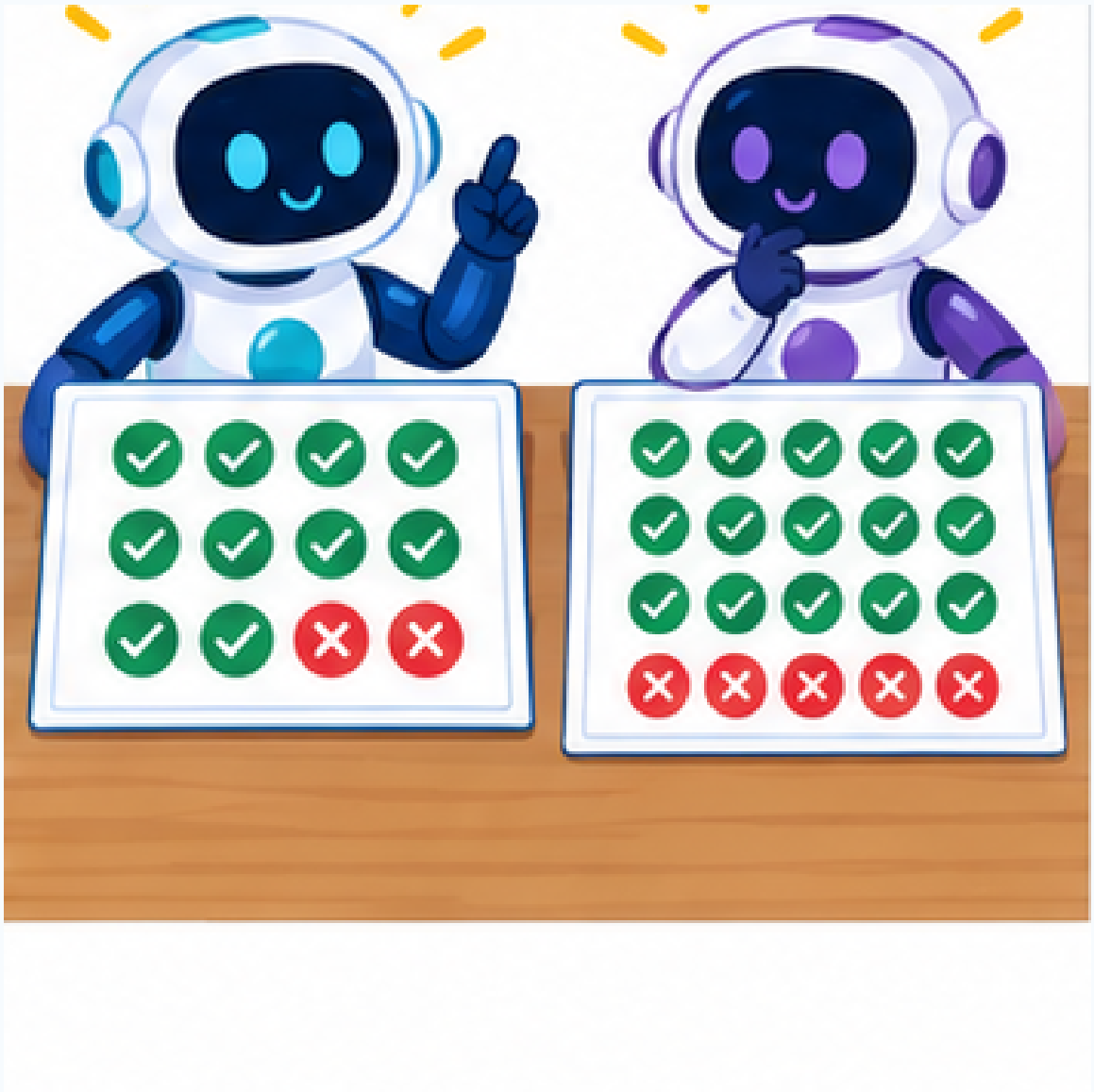
The candidate finds all four but falsely flags four clear cases. The same score conceals a different tradeoff.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **accuracy** as a fraction to solve a problem. What information do they have? What decision or calculation do they make? What would **count** as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Apply the connected idea: Summaries with counts and proportions

A fictional poll asks whether a short activity was useful. Responses are Yes, No or Unknown. We will report Yes among known responses and also disclose unknown responses.

Choosing a **denominator** changes the question being answered.

MAIN LESSON · SECTION 3

Compute ordinary and equal-class summaries separately

> Neither calculation changes a prediction.

Understand the nested lists and returned pair

Read the Python table as two lists: counts[0] is the clear-reference row and counts[1] is the blocked-reference row. Within each row, column 0 means predicted clear and column 1 means predicted blocked. sum adds a row's counts. The function returns ordinary accuracy and the equal average of the two class fractions; the caller unpacks that pair. round affects only displayed percentages.

Predict and run the complete score script

```
def scores(counts):
    clear_n = sum(counts[0])
    blocked_n = sum(counts[1])
    if clear_n == 0 or blocked_n == 0:
        raise ValueError("Need evaluation examples of both classes")
    correct = counts[0][0] + counts[1][1]
    accuracy = correct / (clear_n + blocked_n)
    clear_fraction = counts[0][0] / clear_n
    blocked_fraction = counts[1][1] / blocked_n
    balanced = (clear_fraction + blocked_fraction) / 2
    return accuracy, balanced

baseline = [[16, 0], [4, 0]]
candidate = [[12, 4], [0, 4]]
for name, counts in [("baseline", baseline), ("candidate", candidate)]:
    accuracy, balanced = scores(counts)
    print(name, round(100 * accuracy, 1), round(100 * balanced, 1))
```

Explain the candidate’s two denominators

The **output** is baseline 80.0 50.0, then candidate 80.0 87.5. In each **output** line the first **percentage** is ordinary **accuracy** and the second is **balanced accuracy**.

For the candidate, the clear-class correct fraction is 12/16=75 percent and the blocked-class fraction is 4/4=100 percent. Their equal average is 87.5 percent.

Check assumptions and an absent reference class

For the baseline, those fractions are 16/16=100 percent and 0/4=0 percent, averaging 50 percent. The code assumes a two-by-two table of nonnegative integer **count**s in the stated order.

It stops when either reference class is absent. Then report the observed **count**s and the unavailable class result; do not invent a **denominator** or silently substitute zero.

Connect a picture to a denominator

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Optional denominator extension: the complete poll script

```
responses = ["Yes", "No", "Yes", "Unknown", "Yes"]
yes = 0
known = 0
for response in responses:
    if response == "Yes" or response == "No":
        known = known + 1
        if response == "Yes":
            yes = yes + 1
if known > 0:
    print(yes / known)
else:
    print("No known responses")
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Yes	3	Numerator
Yes or No	4	Known denominator
All supplied	5	Includes one unknown

Read each row from left to right and connect the evidence to the stated rule.

The printed **proportion** is 0.75, or 75 percent, among four known responses. Yes among all five supplied responses is 0.6, or 60 percent. Both calculations are correct when labelled; they describe different **denominator**s.

Unknown must not silently become No.

Count the responses before dividing. In the example, three say Yes, one says No and one is Unknown. There are four known responses, so Yes among known responses is 3 divided by 4, or 75 percent.

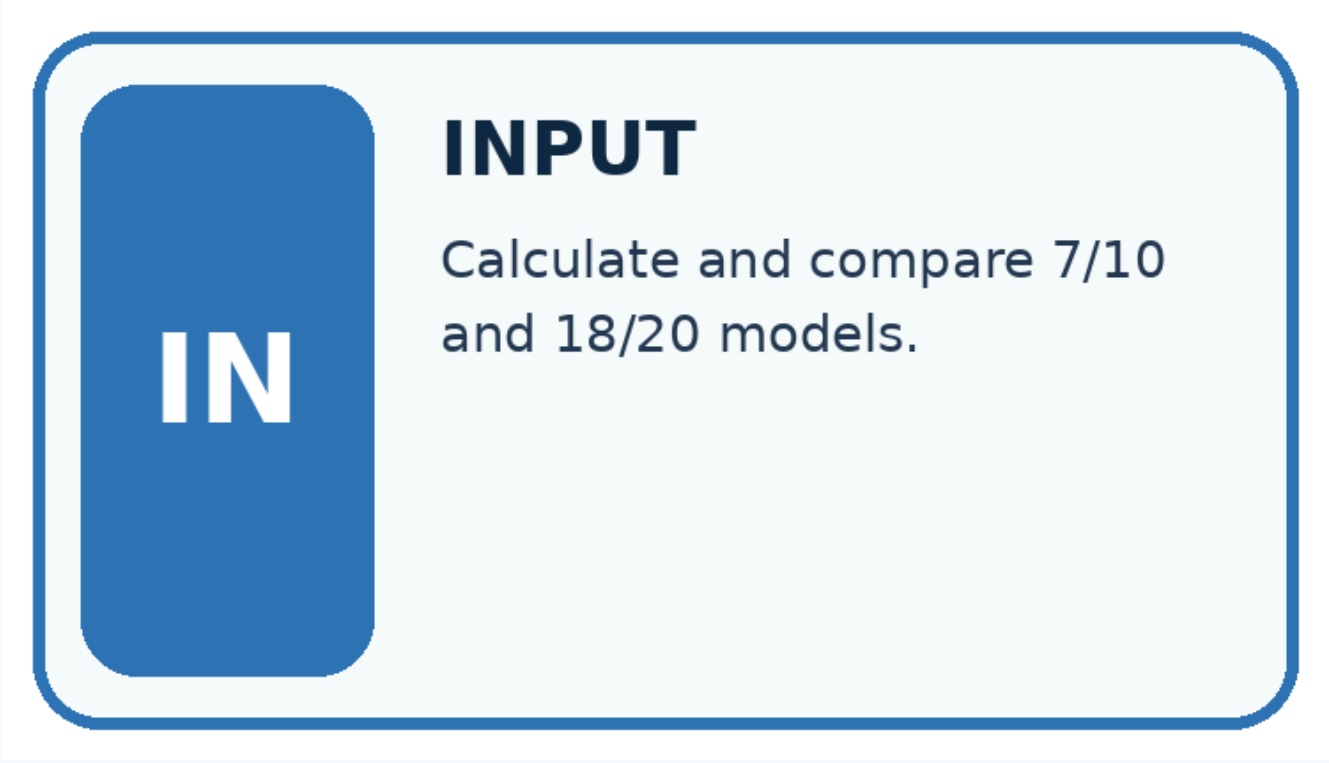
Yes among all five supplied responses is 3 divided by 5, or 60 percent. Neither number needs to be hidden. Label the question each answers, and keep the unknown **count** visible.

▼ **Earlier wording, alternatives and picture transcripts**

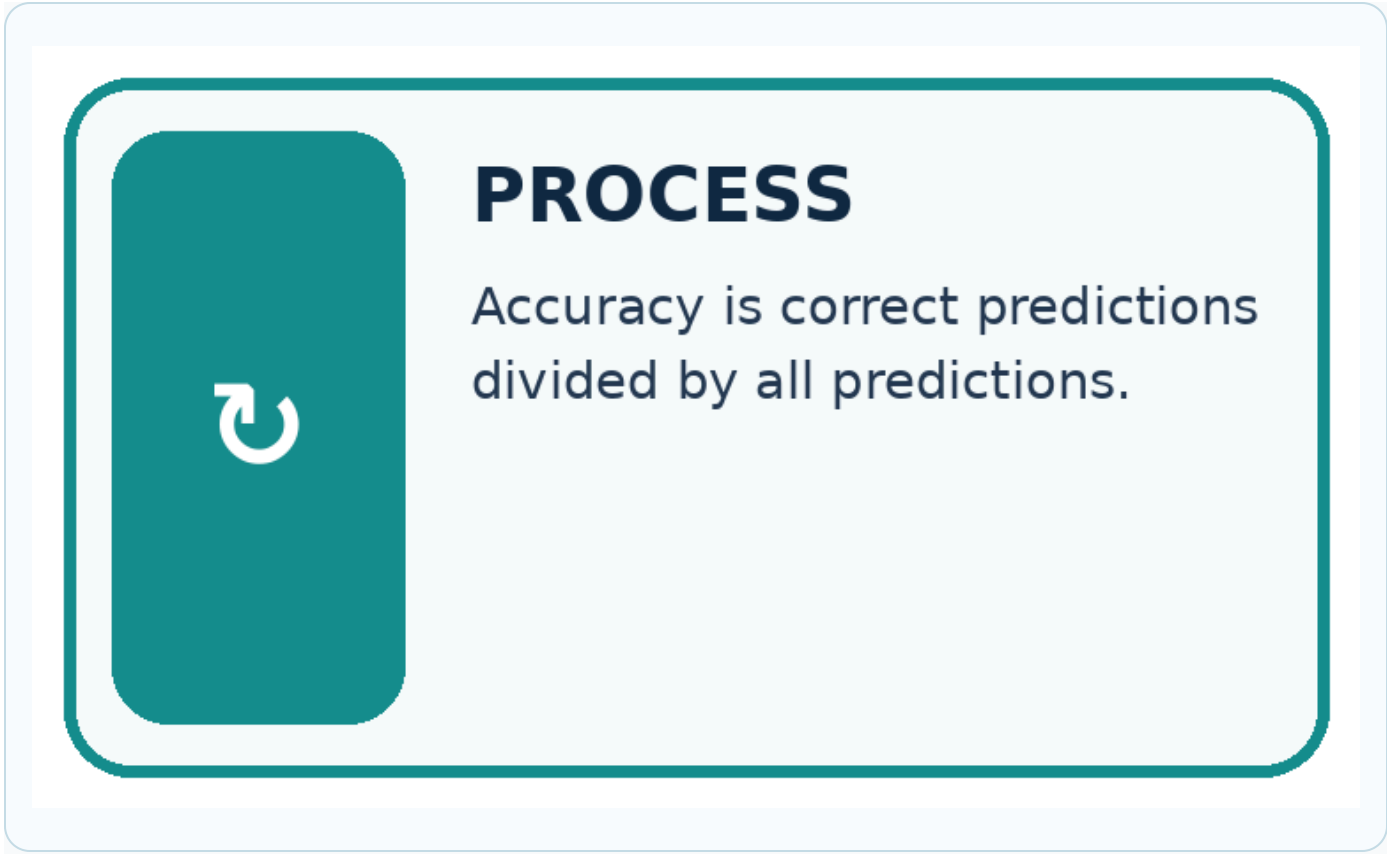
VISUAL **MECHANISM**

Input → **Process** → **Output**

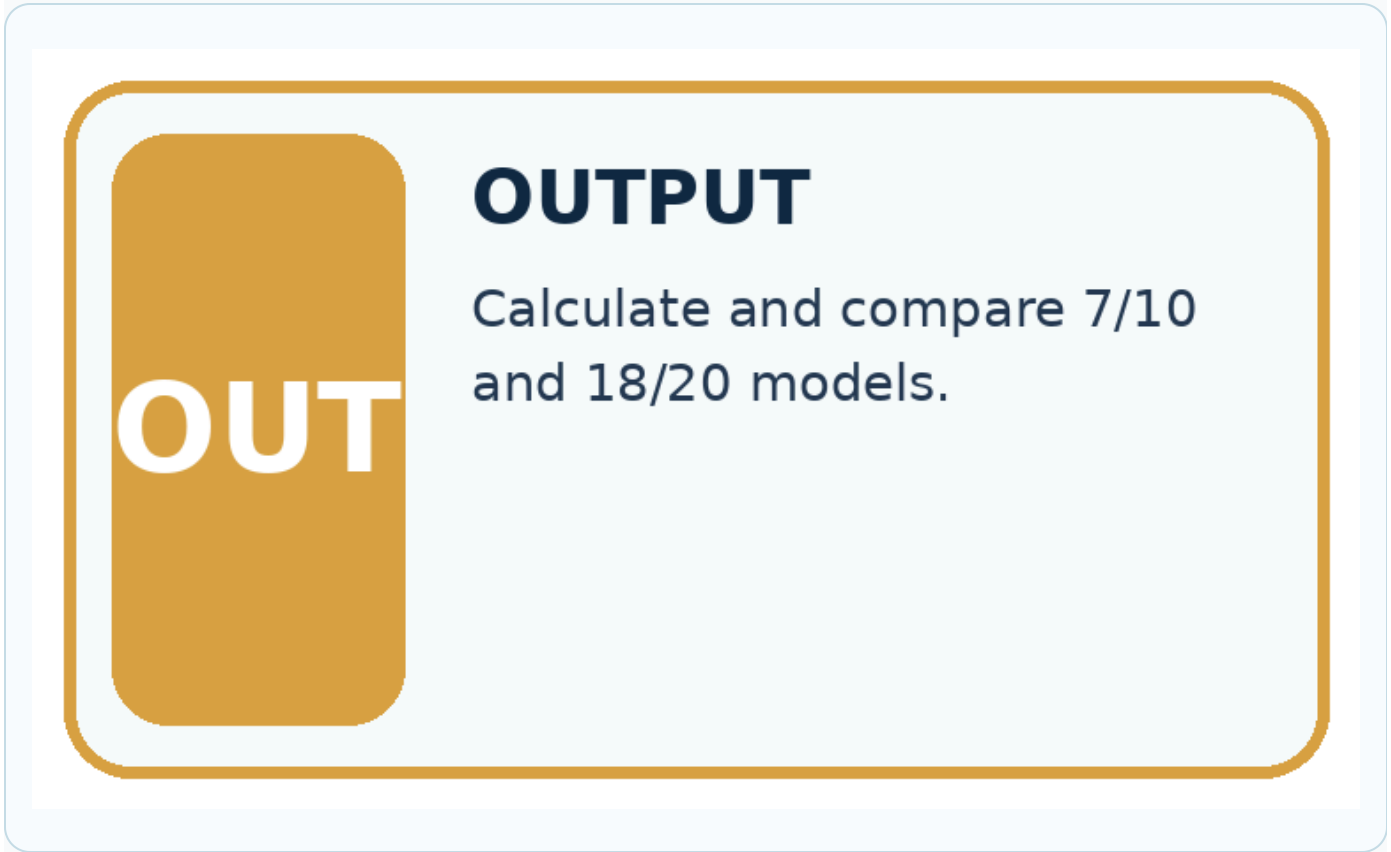
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Accuracy is correct predictions divided by all predictions.

↓


OUT
OUTPUT
Calculate and compare 7/10 and 18/20 models.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: Summaries with counts and proportions

Name the group behind a percentage

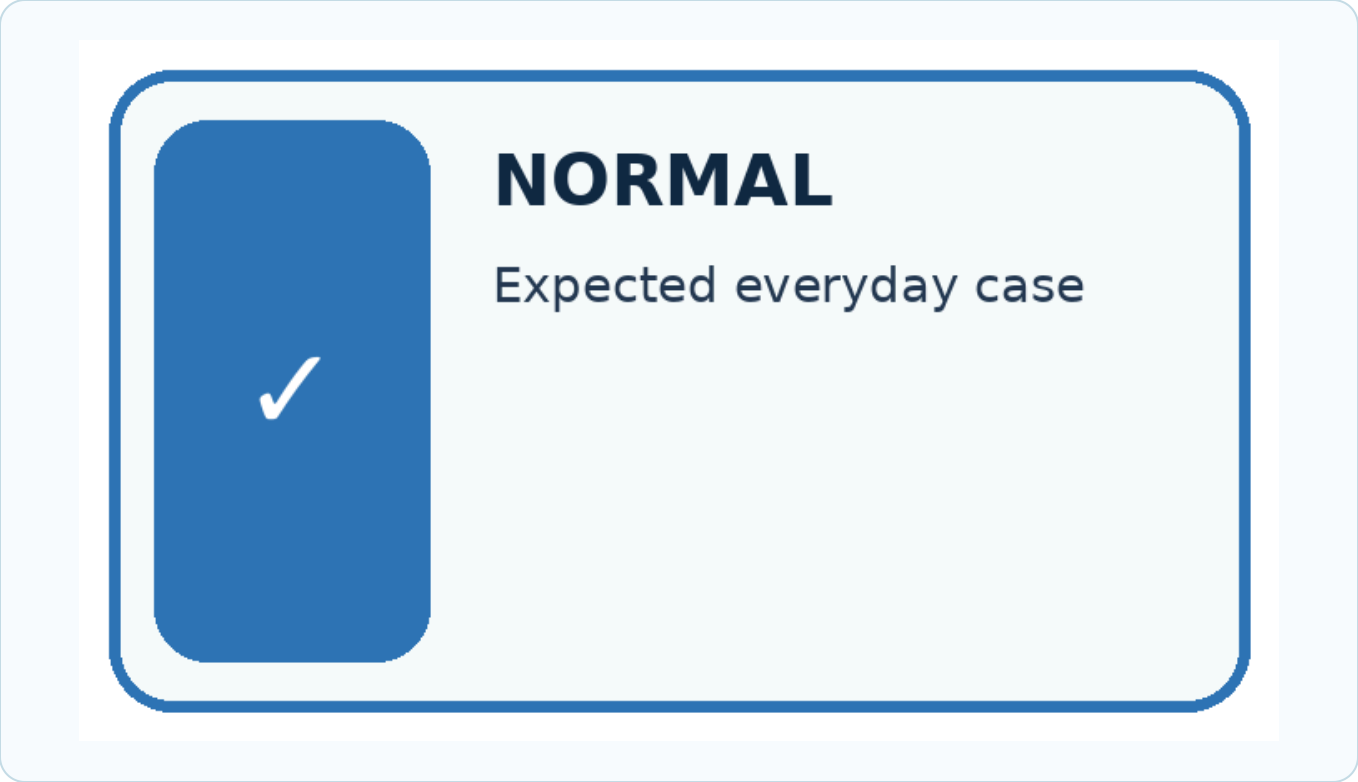
Connect the picture and the explanation

MAIN LESSON · SECTION 4

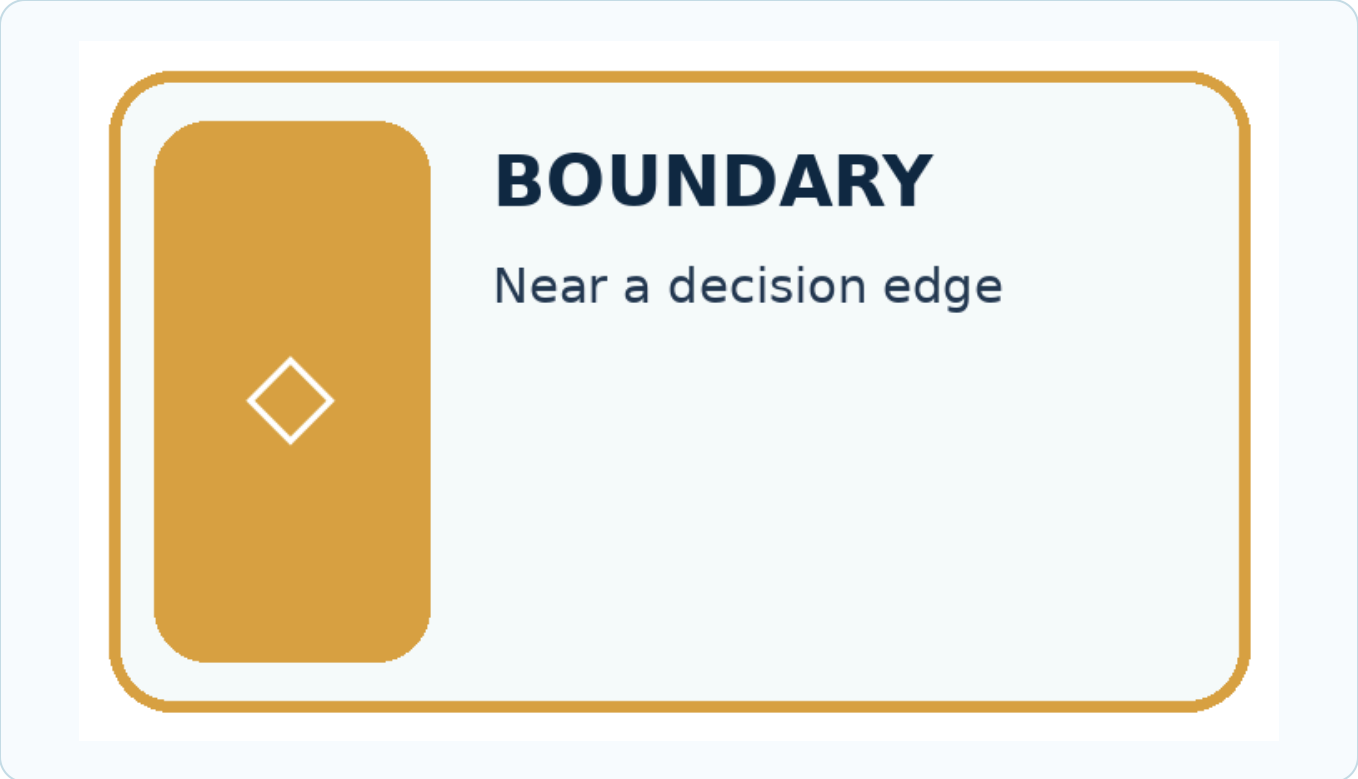
Challenge the headline score

> Ask which denominator and which class the result describes.

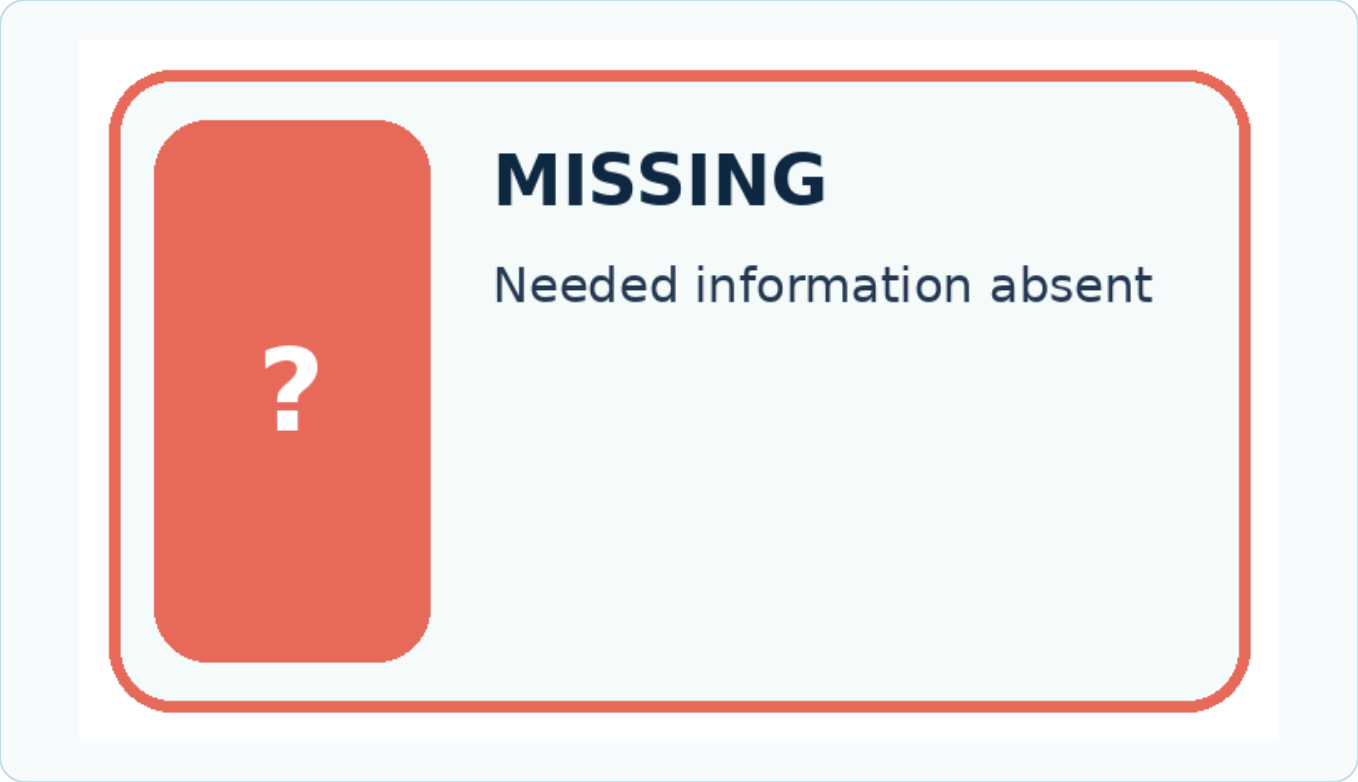
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: A candidate instead marks every book repair needed.

▼ Check after predicting

Prediction: Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

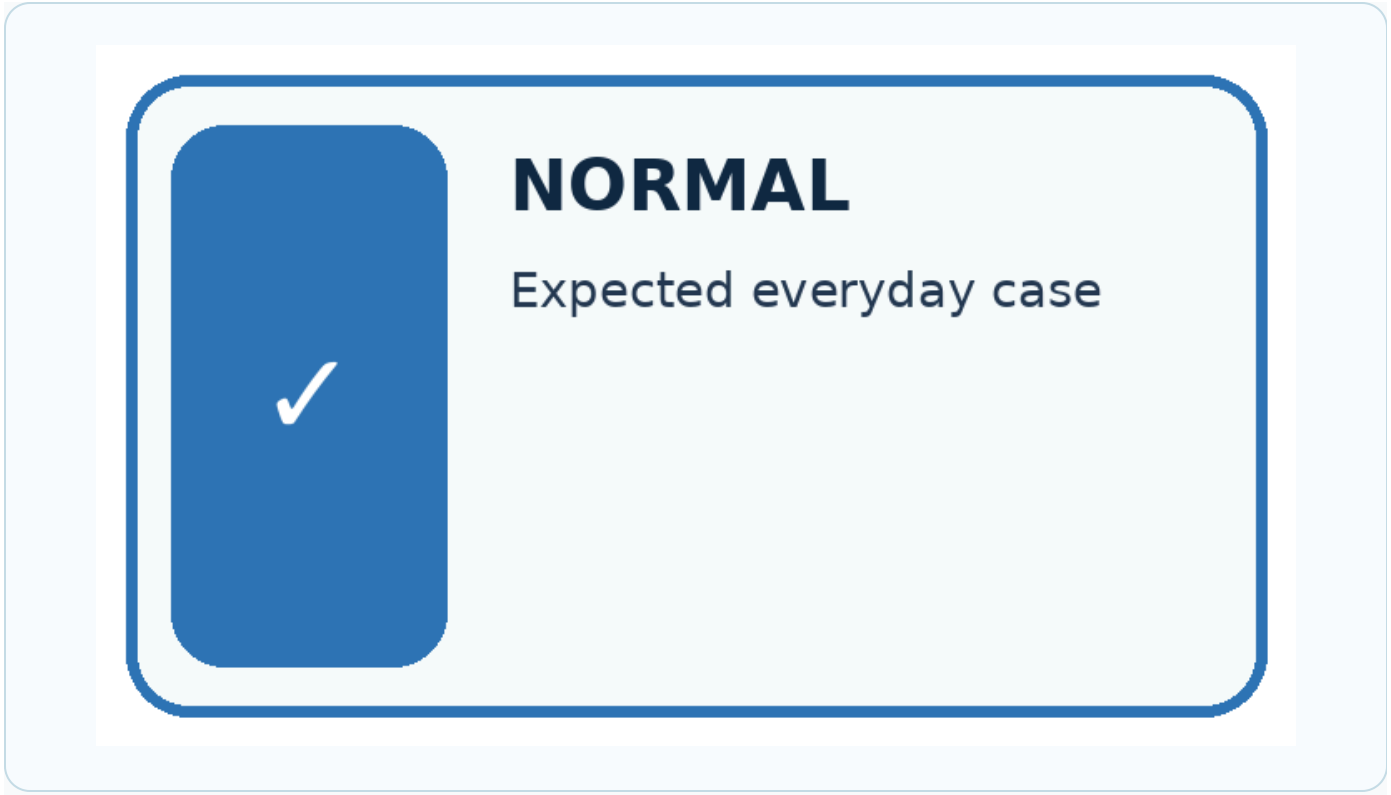
Reason: All 16 not-needed books become false alarms.

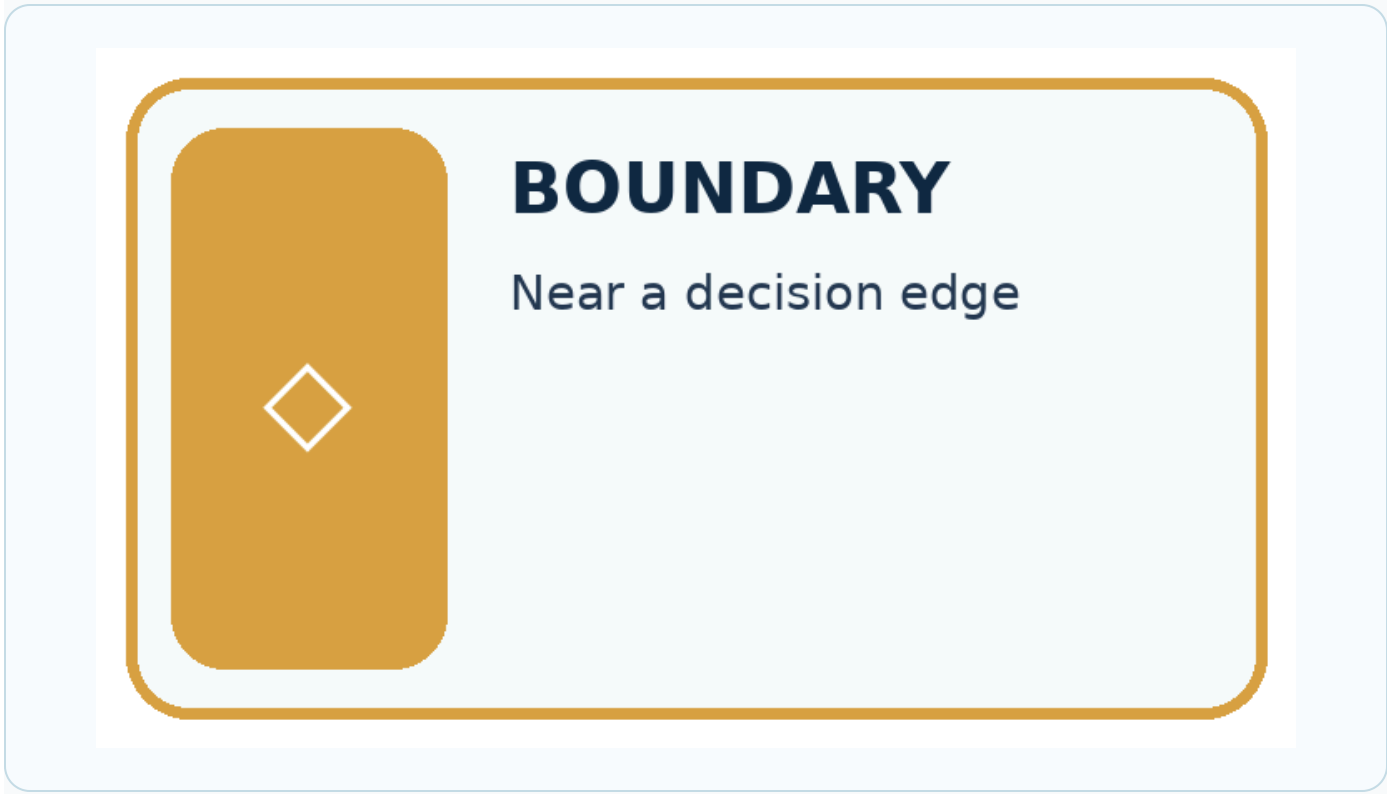
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Explain the equal-score case before a separate shelf example

> The shelf candidate has different counts; do not mix its 85% with the main candidate's 80%.

Compare the main methods class by class

Same overall score different class results

SUMMARY	ALWAYS CLEAR BASELINE	FROZEN CANDIDATE
Overall correct	16 of 20 is 80 percent	16 of 20 is 80 percent
Reference clear correct	16 of 16 is 100 percent	12 of 16 is 75 percent
Reference blocked correct	0 of 4 is 0 percent	4 of 4 is 100 percent

Overall accuracy weights classes by their number of evaluation examples. Balanced accuracy gives the two class-specific fractions equal weight.

Explain their equal ordinary accuracy

> **Both methods score 80 percent. Explain which cases the baseline fails, how the candidate changes the errors, and why their balanced accuracies differ.**

Limit what balanced accuracy establishes

Balanced accuracy is a metric choice, not a repair to the classifier or a guarantee of safety. It gives classes equal influence in this summary; it does not encode the actual consequences of false alarms and missed cases.

Four blocked evaluation examples are also too few for a strong general claim.

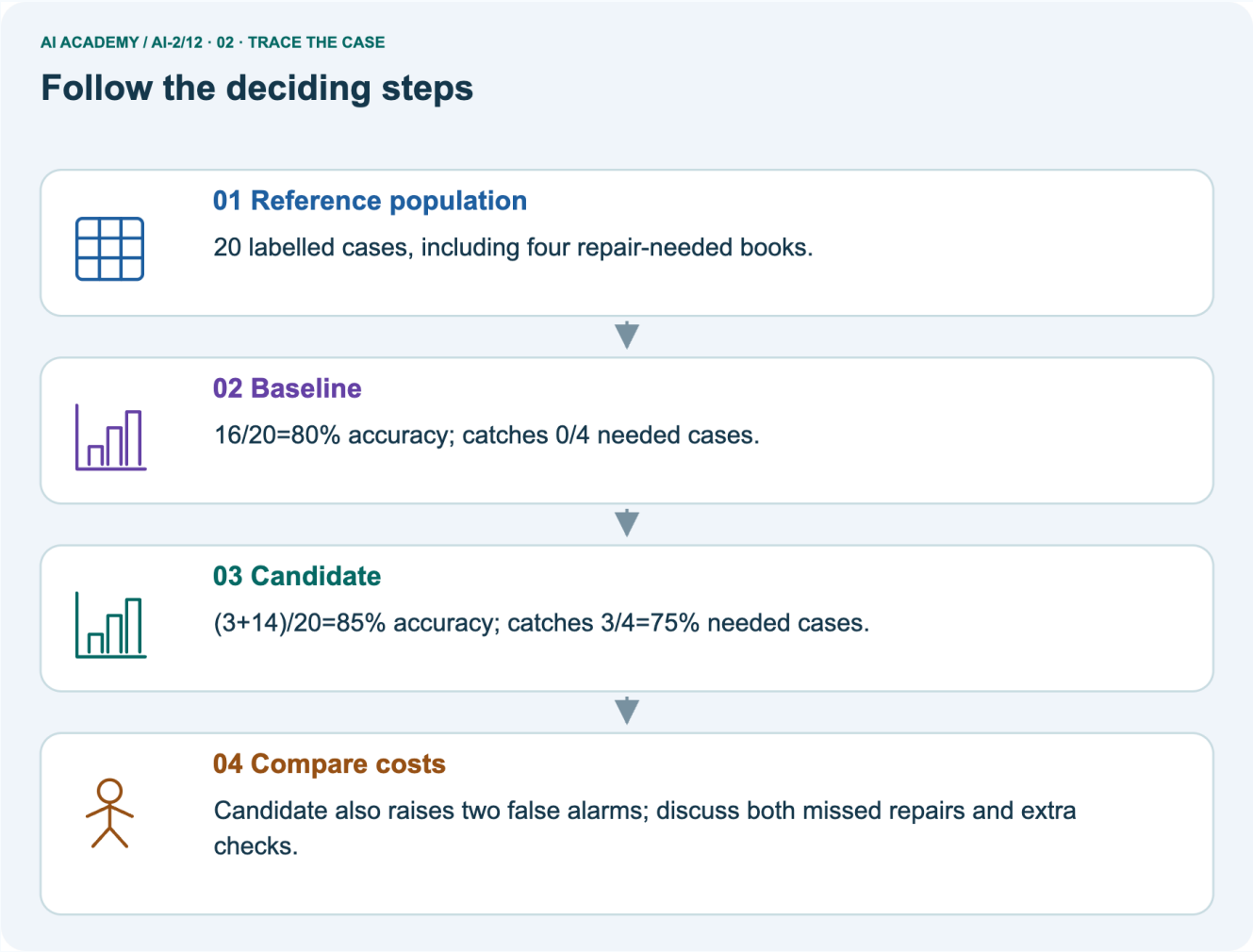
Introduce the separate shelf-repair evaluation

A fictional shelf classifier predicts repair needed or not needed.

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Always-not-needed baseline is correct on 16. Candidate has TP=3,FN=1,FP=2,TN=14.

Trace missed repairs and false alarms



Follow the labelled picture with the matching explanation. Open the vector diagram

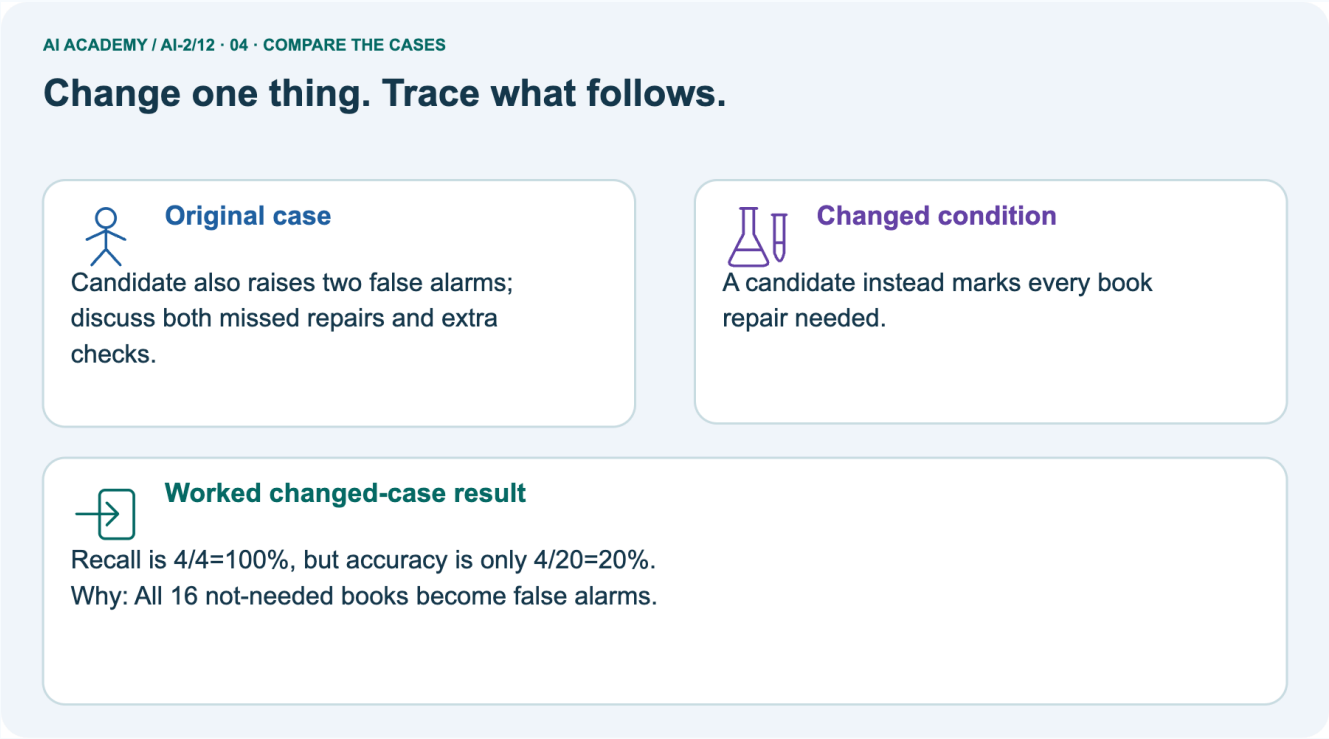
A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Predict the always-repair-needed change

- Change one thing: A candidate instead marks every book repair needed. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the changed tradeoff and evidence limit

- Read the labelled picture: Why is 100% repair detection not sufficient to declare the changed classifier best?
- These selected cases do not establish future rates or specify the real cost of each error.

▼ Optional poll denominator question

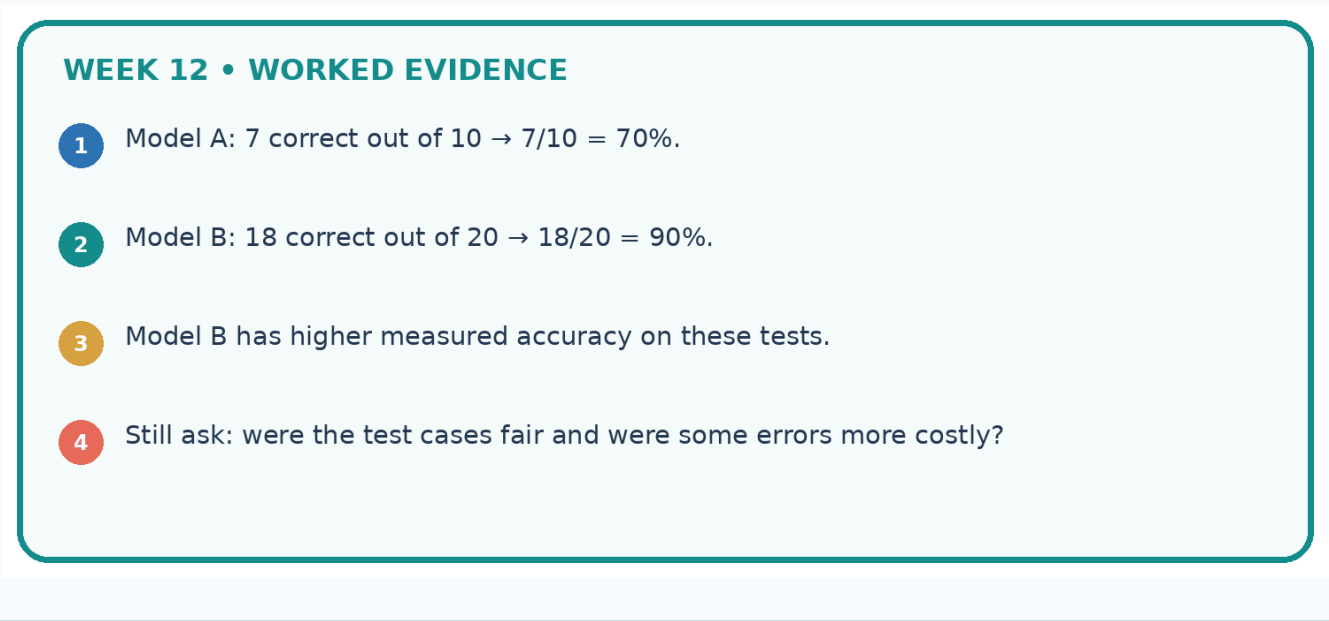
> **For Yes, No, Unknown, Unknown, calculate Yes among known and Yes among all supplied.**

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Model A: 7 correct out of 10 → 7/10 = **70%**.

2. Model B: 18 correct out of 20 → 18/20 = 90%.

3. Model B has higher measured **accuracy** on these tests.

4. Still ask: were the test cases fair and were some errors more costly?

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

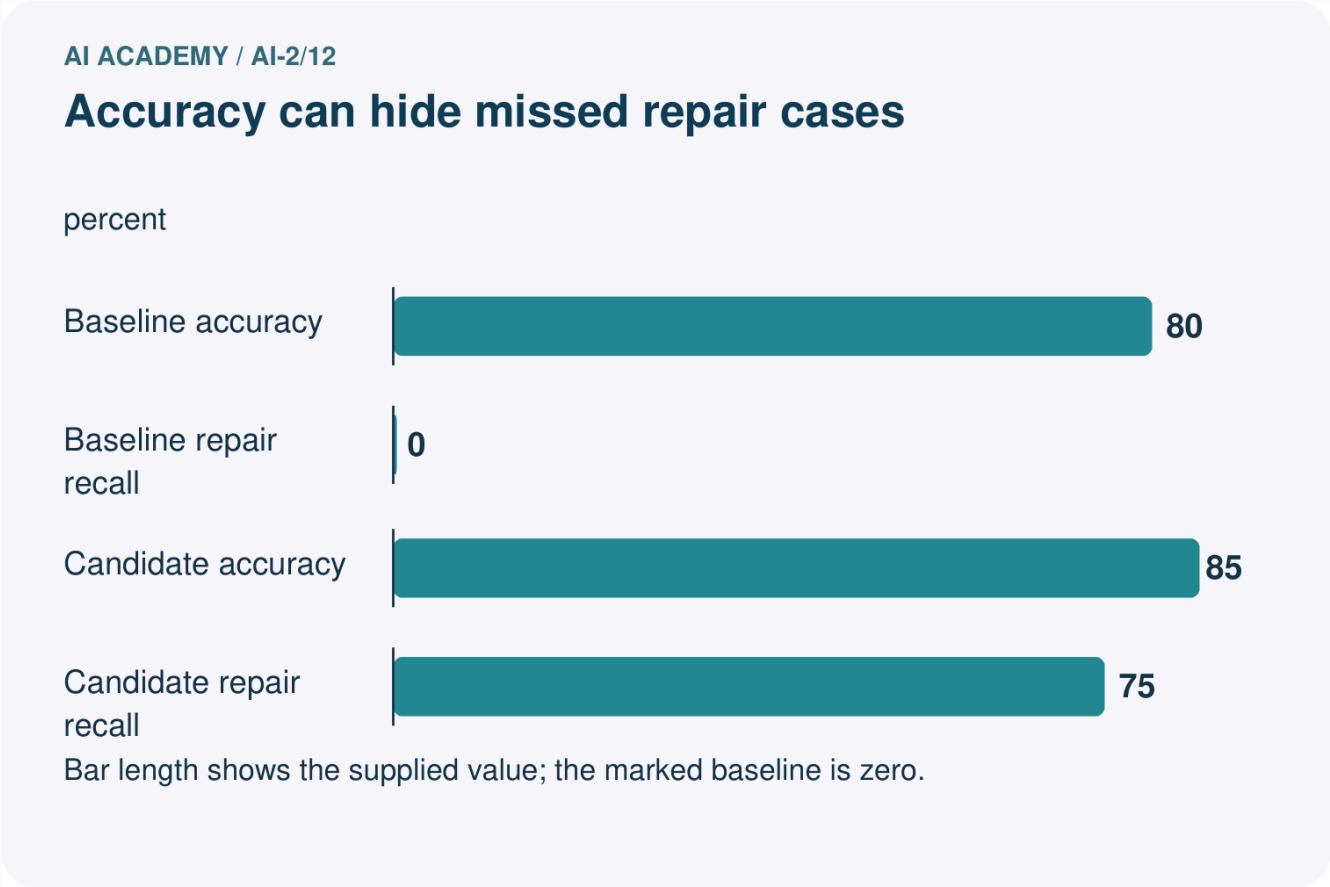
Apply the connected idea: Summaries with counts and proportions

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Accuracy as a Fraction

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/12

Accuracy as a Fraction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reference population

20 labelled cases, including four repair-needed books.



2. Baseline

16/20=80% accuracy; catches 0/4 needed cases.



3. Candidate

(3+14)/20=85% accuracy; catches 3/4=75% needed cases.



4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Reference population	20 labelled cases, including four repair-needed books.
2. Baseline	16/20=80% accuracy; catches 0/4 needed cases.
3. Candidate	(3+14)/20=85% accuracy; catches 3/4=75% needed cases.
4. Compare costs	Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Ask for counts and coverage beside a percentage

> Do not turn four observed successes into a universal guarantee.

State the fair comparison and its limits

Compare methods on the same appropriate evaluation examples and report the **training**-derived baseline. Include ordinary **accuracy**, each class **denominator**, false positives and false negatives.

An equal-class summary adds information but cannot choose the acceptable tradeoff for the task. Keep development decisions separate from a final test.

A result of 4/4 correct blocked cases is 100 percent on those four examples, not proof of perfect future detection. If the evaluated set contains no blocked cases, it cannot measure blocked-case performance at all. Next is Week13, Types of Classification Error. Carry forward the positive class, matrix orientation and distinct false-alarm and missed-case counts; later rate summaries must name their **denominator**s.

Challenge an always-clear headline

> **A team advertises 80 percent accuracy but its classifier predicts clear on every record. What additional evidence would you request, and what can you already conclude about the four blocked cases in our worked evaluation?**

▼ Optional poll: volunteers and representativeness

A **percentage** from a few volunteers is not automatically representative of the whole class. Report **count**s, the question, who responded and exclusions alongside the **percentage**.

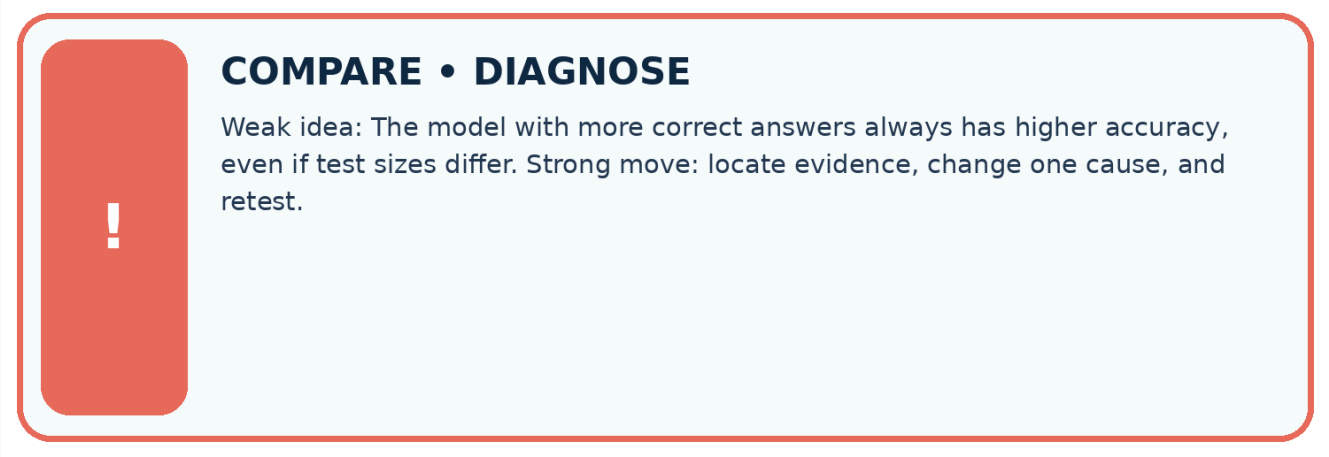
> **Three of five known volunteers said Yes. Can you say 60 percent of all students agree?**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Apply the connected idea: Summaries with counts and proportions

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MAIN LESSON · SECTION 8

Change the class mix without changing the assumed class fractions

> Treat the comparison as arithmetic, not an observed population experiment.

The separate eight-and-eight calculation

Consider a separate illustrative set with 8 clear and 8 blocked cases. Suppose the same within-class fractions hold: the baseline matrix is [[8,0],[8,0]] and the candidate matrix is [[6,2],[0,8]].

Ordinary **accuracy** is now 50 percent versus 87.5 percent. **Balanced accuracy** remains 50 percent versus 87.5 percent. The class mix changed the overall score even though the within-class fractions stayed fixed.

State what changed and what remains unproven

This is an arithmetic comparison, not proof that a real classifier keeps the same behavior after a population change. Report sample size, class **count**s, full matrix and how the evaluation data was chosen.

Do not remove inconvenient cases or duplicate rare evaluation cases just to improve a headline number. If **training** is resampled, keep that separate from evaluation and document it.

Practise the main candidate in W2

Use workbook W2 for the connected example "**Accuracy** and **class imbalance**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair high accuracy means both classes work

- Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the shelf false-alarm tradeoff

Why is 100% repair detection not sufficient to declare the changed classifier best?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Optional poll: zero denominator and further practice

Dividing by known when it equals 0 causes ZeroDivisionError. Test known > 0 before division and report no known responses if none exist. Reporting 0 percent would wrongly suggest known responses with no Yes answers.

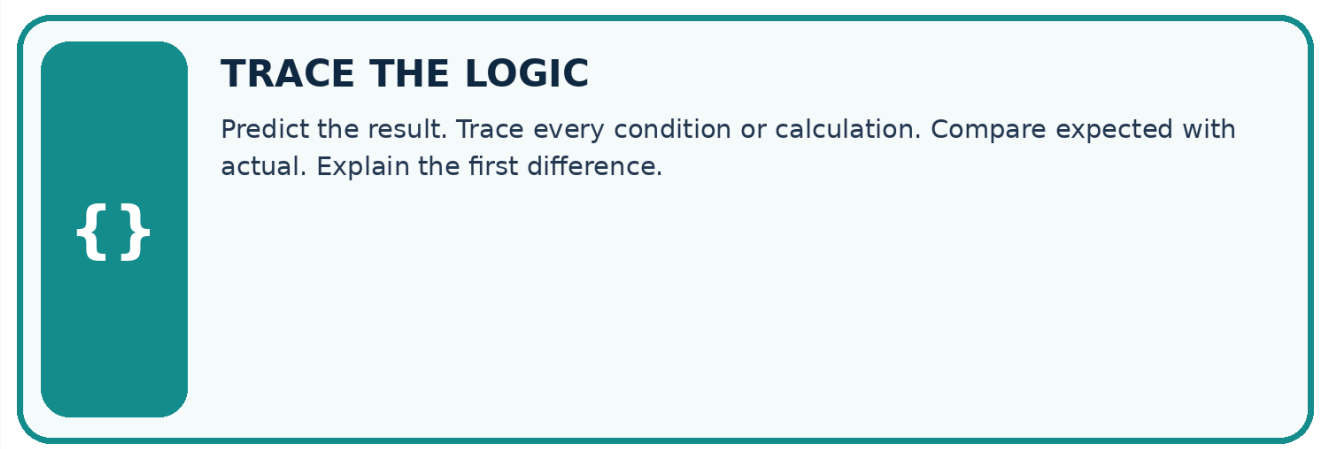
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Apply the connected idea: Summaries with counts and proportions

Practise with less help: Compare both class fractions

Check your reasoning

Practise the connected topic: Summaries with counts and proportions

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Rebuild two matrices from supplied prediction sheets

> Keep raw cases, counts, denominators and error types linked.

The frozen rules and evaluation scope

Use these supplied, frozen prediction records for the same 20 fictional evaluation cases. Training has 16 clear and 4 blocked labels, so the baseline was chosen as always-clear before this evaluation. In the table, an ID range represents every individual case in that range, with the listed reference and predictions. No classifier is trained or changed during this arithmetic audit.

The complete twenty-case prediction record

Case IDs · Count · Reference · Baseline prediction · Candidate prediction

> **Case IDs:** 1–12
>
> **Count:** 12
>
> **Reference:** clear
>
> **Baseline prediction:** clear
>
> **Candidate prediction:** clear

> **Case IDs:** 13–16
>
> **Count:** 4
>
> **Reference:** clear
>
> **Baseline prediction:** clear
>
> **Candidate prediction:** blocked

> **Case IDs:** 17–20
>
> **Count:** 4
>
> **Reference:** blocked
>
> **Baseline prediction:** clear
>
> **Candidate prediction:** blocked

Predict, tally, calculate and check

1. Before running the score script, tally all four cells for each method from the supplied records. Calculate correct/all, clear-correct/all-clear and blocked-correct/all-blocked, retaining each fraction before its percentage. Average the two class fractions equally for balanced accuracy. Execute the supplied script to check the aggregate arithmetic, or label your check a paper calculation. Record FP and FN separately and explain what tradeoff a human must decide. Then use the separate 8-clear/8-blocked matrices in Section 8; do not describe those assumed counts as new observed model behaviour. Reserve the 18-clear/2-blocked case for independent W4.

Keep the evidence and explain the revealing case

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

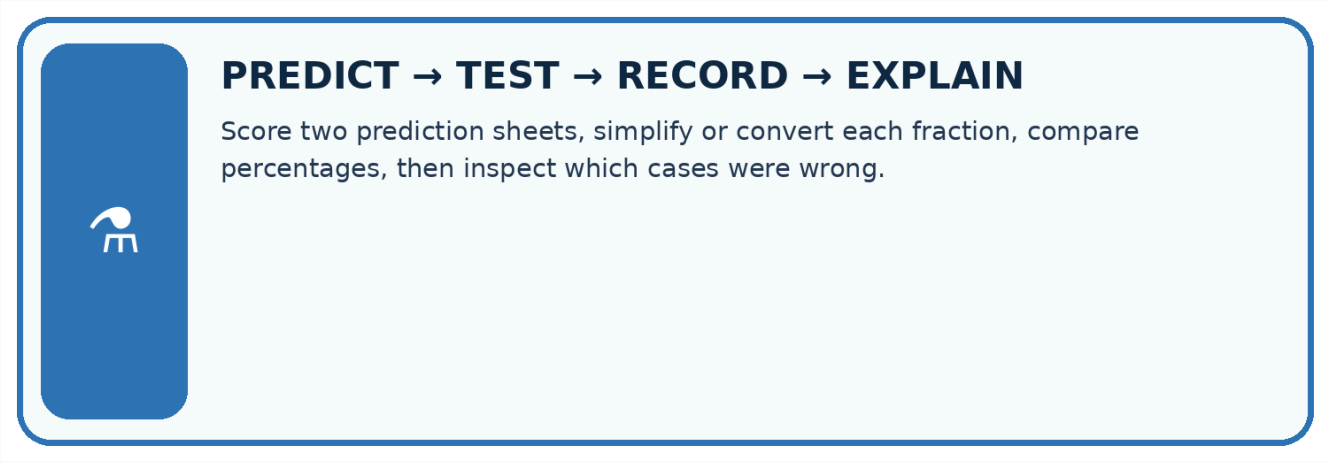
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied twenty-case **prediction** record to tally both matrices, calculate ordinary and balanced accuracy, and inspect false alarms and missed cases. Keep the independently assessed 18-clear/2-blocked case unused during shared practice.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare methods on the new 18-and-2 evaluation

> Preserve every denominator and explain the tradeoff before feedback.

The reserved matrices and question

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Assess a new comparison independently

Week 12 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Report correct recommendations / all test users, alongside the raw test **count**.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Define a scoreable recommender task before reporting accuracy

> A denominator needs a declared unit and reference label.

This week’s portfolio contribution

Before reporting accuracy for the fictional recommender, define one evaluated case, its frozen target label and what counts as a correct prediction. A preference ranking or enjoyable recommendation has no automatic correct label. Use a declared, source-supported classification task for this week’s score, compare with a training-derived baseline on the same cases, report class counts and retain uncertain or unscored cases separately. Use invented profiles rather than real learner data.

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

Keep the system evidence and human decision visible

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

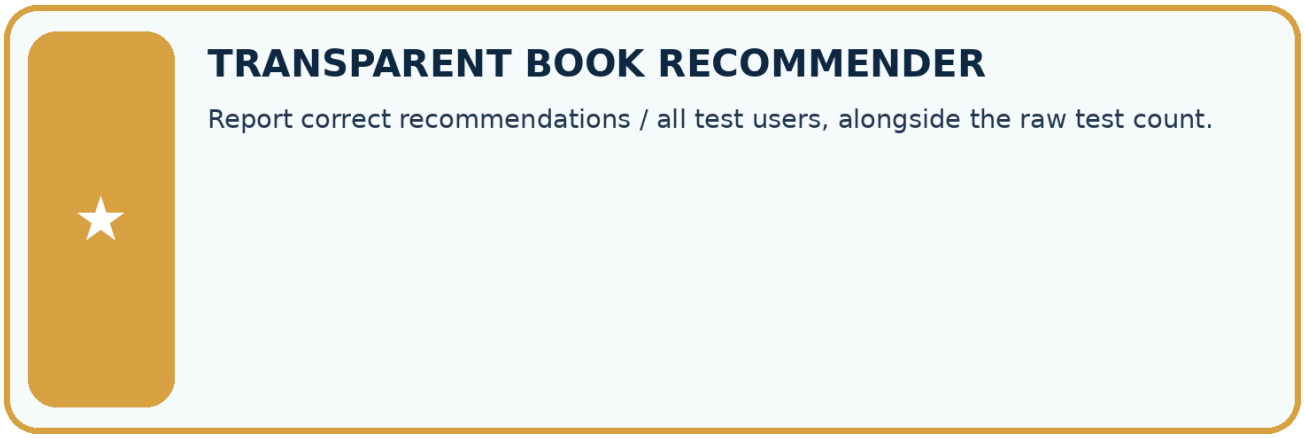
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Report correct recommendations / all test users, alongside the raw test **count**.

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain what the accuracy denominator hides

> Give a class-specific count and a remaining limit.

Your explanation without notes

- Close your notes. Explain this goal in your own words: Explain why overall **accuracy** can hide failure on a less common class

My deciding step, evidence and one remaining question

Return on a changed case

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for scores and denominators**

REFERENCE · SECTION 5

Reference · Words for scores and denominators

> Use each term with a concrete count.

Core words

Word	My own example	Not this / boundary
accuracy	The number of correct predictions divided by the total number of scored predictions.	
numerator	The top number of a fraction, identifying the counted portion.	
denominator	The bottom number of a fraction, identifying the number of equal parts or total cases considered.	
percentage	A proportion expressed per hundred; multiply a fraction by 100 to express it as a percent.	

Class-imbalance vocabulary

Word	Meaning
Class imbalance	Unequal numbers of examples in different classes

Word	Meaning
Majority baseline	A rule that always predicts the most common training label
Class-specific correct fraction	Correct predictions for one reference class divided by the number of examples of that class
Balanced accuracy	The average of the class-specific correct fractions with equal weight for each class

Poll-extension vocabulary

Term	Meaning
Count	The number of matching records
Denominator	The total number used in a fraction
Proportion	A part divided by its relevant whole
Eligible	Included under a stated rule

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

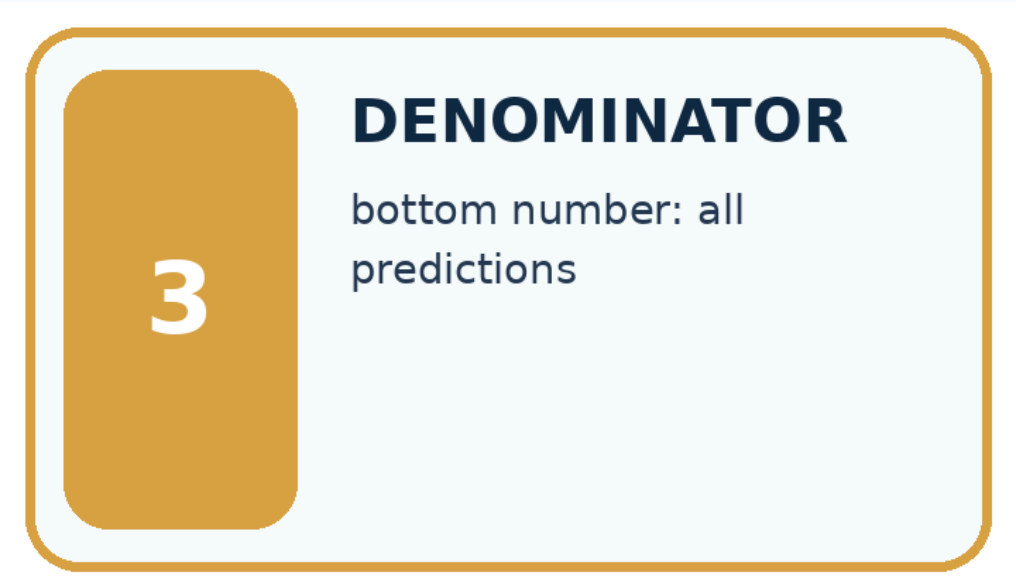
ACCURACY

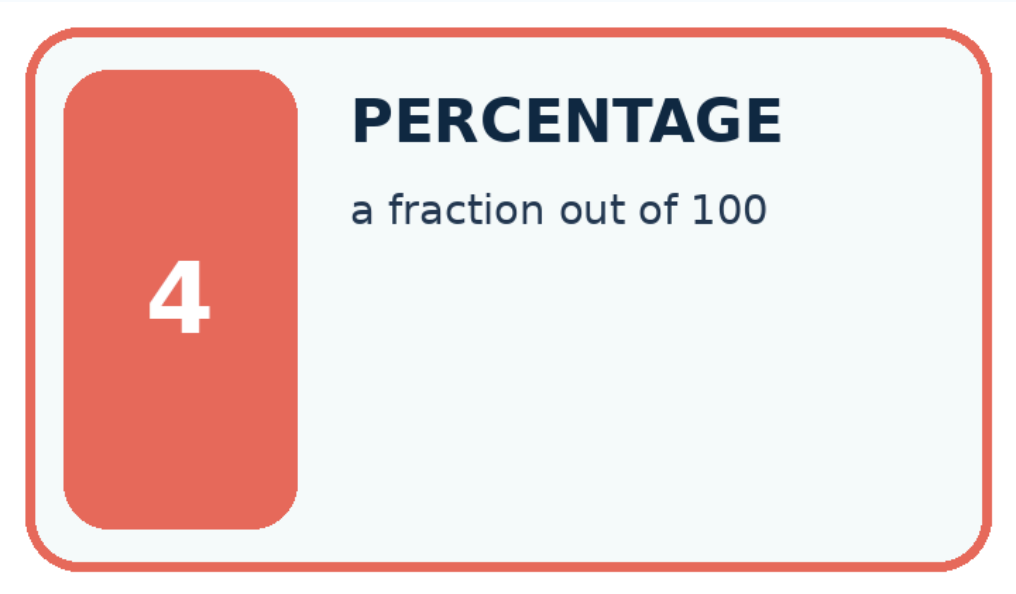
correct predictions divided by all predictions

2

NUMERATOR

top number: correct predictions





Words used in the closer explanation

Terms for Summaries with counts and proportions

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Select the next useful question rather than repeating every prompt.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, **count**erexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

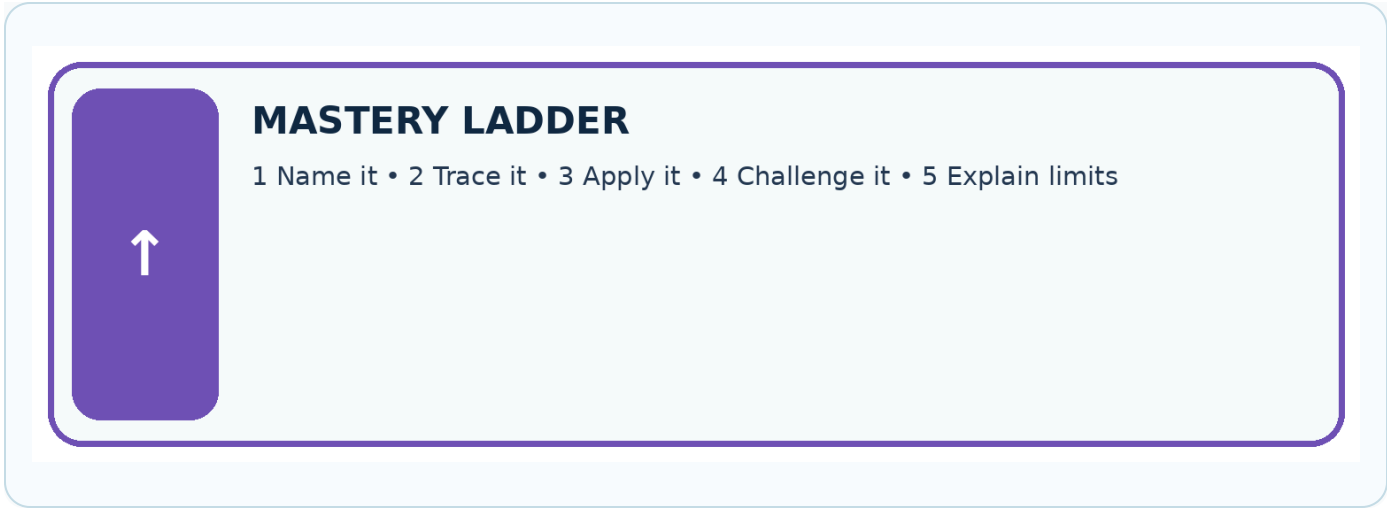
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation after practice

REFERENCE · SECTION 12

Reference · Check your explanation after practice

> Use the selected 18-and-2 case for the independent first attempt.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Record the check

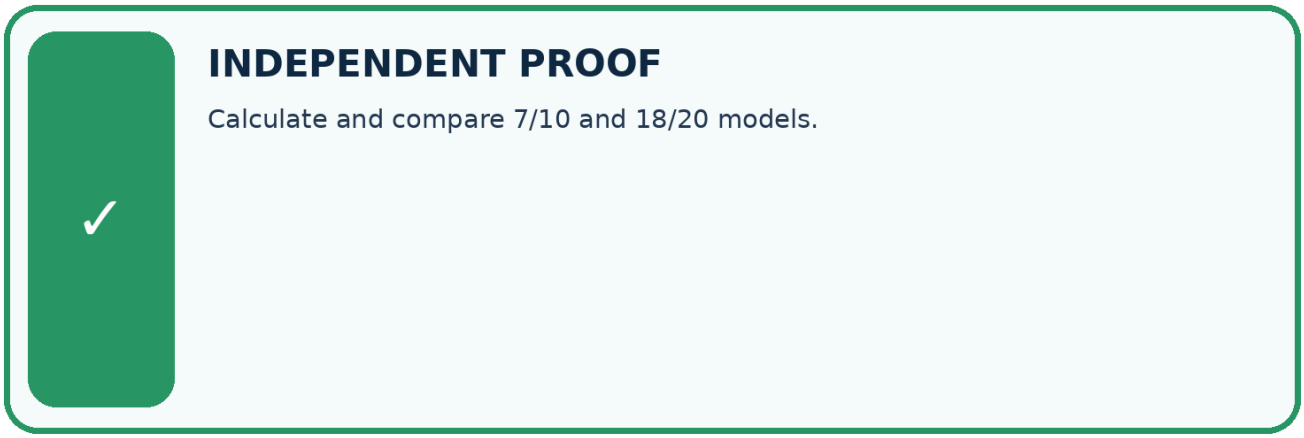
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining scores and denominators

Use after practice. The selected18-clear/2-blocked case is the independent first attempt.



Explain ordinary and balanced accuracy with class denominators, error counts and a limit on the conclusion.

▼ Extra practice · A simple accuracy denominator

REFERENCE · SECTION 13

Extra practice · A simple accuracy denominator

> Try 9 out of 12 before checking the worked result.

The question

Nine of twelve held-out classifications are correct.

Compute **accuracy** and state the **denominator**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

9/12=3/4=75%. **Count** all scored cases, including errors.

Evidence or stage	Value or result
Correct classifications	9/12

▼ Earlier wording, alternatives and picture transcripts

Week 12 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

All scored cases belong in the denominator

Nine of twelve held-out classifications are correct.

Compute **accuracy** and state the **denominator**.

9/12=3/4=75%. **Count** all scored cases, including errors.

▼ Reference · Earlier target checklist

REFERENCE · SECTION 15

Reference · Earlier target checklist

> Use only when it helps inspect your evidence.

Targets and first prediction

By the end, I can	How I will show it
Explain why overall accuracy can hide failure on a less common class	Explain the deciding step, show a trace or test, and state one limit.
Compare a training-majority baseline with class-specific results	Explain the deciding step, show a trace or test, and state one limit.
Calculate an equal-class average and report its limits alongside the confusion matrix	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional shelf classifier predicts repair needed or not needed.

▼ Extra practice · Pictured evidence and repair

REFERENCE · SECTION 17

Extra practice · Pictured evidence and repair

> Try before opening feedback; keep independent W4 separate.

First question

- A fictional shelf classifier predicts repair needed or not needed.

- Which labelled step matches this detail? Candidate also raises two false alarms; discuss both missed repairs and extra checks.

- 1. Reference population

- 2. Compare costs

- 3. Candidate

- 4. Baseline

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ____
>
> ____
>
> ____
>
> ____

▼ **Check after trying**

After your attempt, compare with the supported explanation: Compare costs Candidate also raises two false alarms; discuss both missed repairs and extra checks.

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Changed-case question

- Changed-case practice: A candidate instead marks every book repair needed. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Recall is 4/4=100%, but **accuracy** is only 4/20=20%.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the change**

After your attempt, compare with the supported explanation: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. All 16 not-needed books become false alarms.

Repair and keep your work

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan **count**s?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
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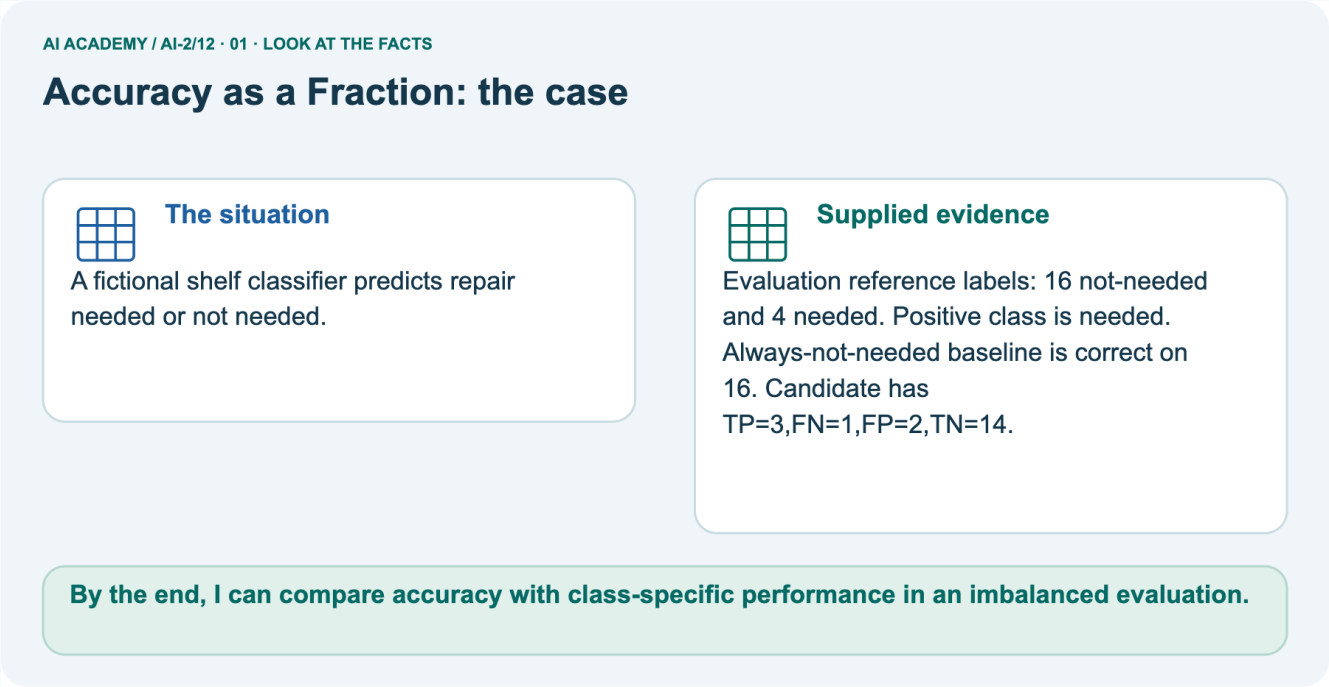
<details>
<summary>Reference · The six-picture journey</summary>
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Accuracy as a Fraction: the case



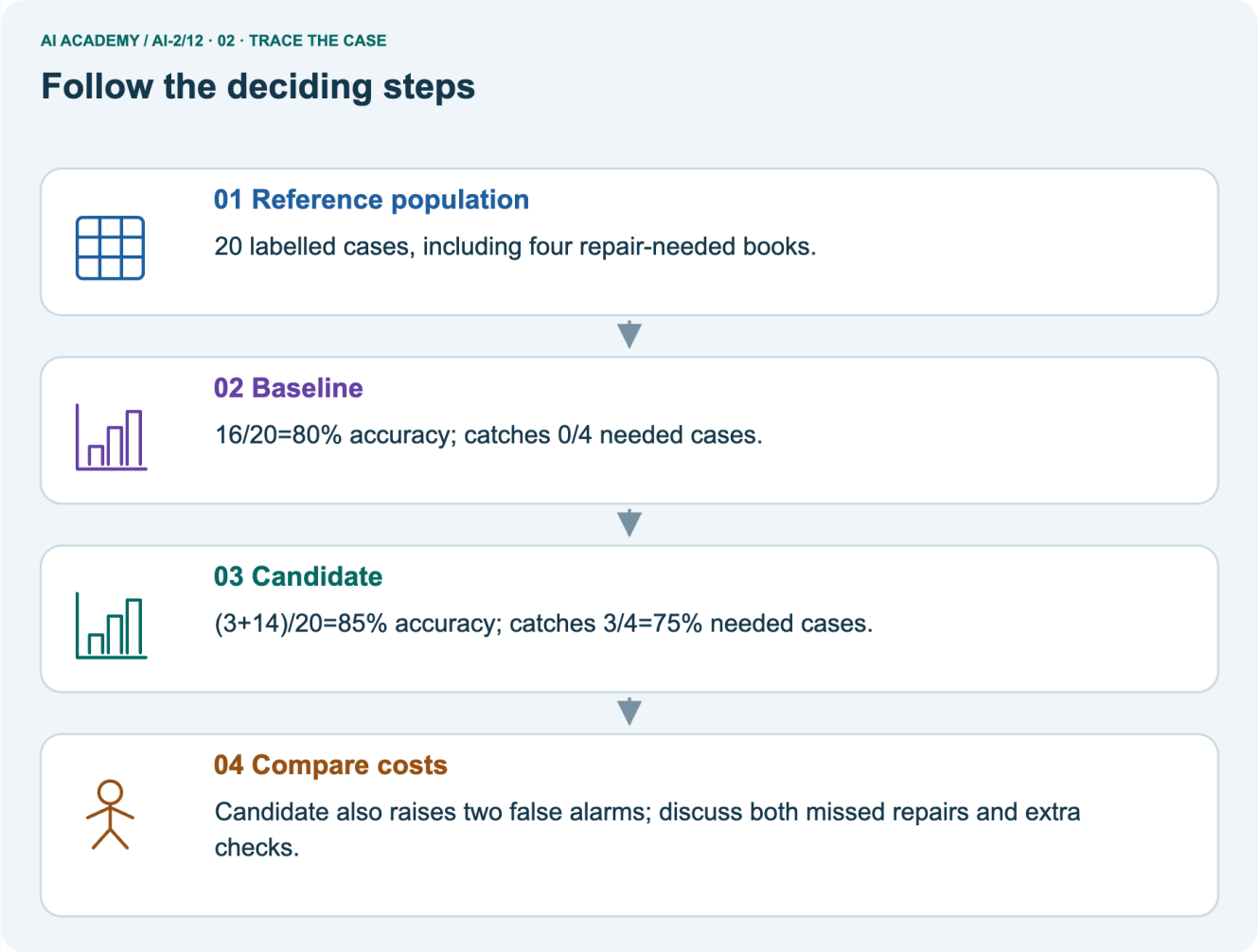
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional shelf classifier predicts repair needed or not needed. Supplied facts: Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed. Always-not-needed baseline is correct on 16.

Candidate has TP=3,FN=1,FP=2,TN=14.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

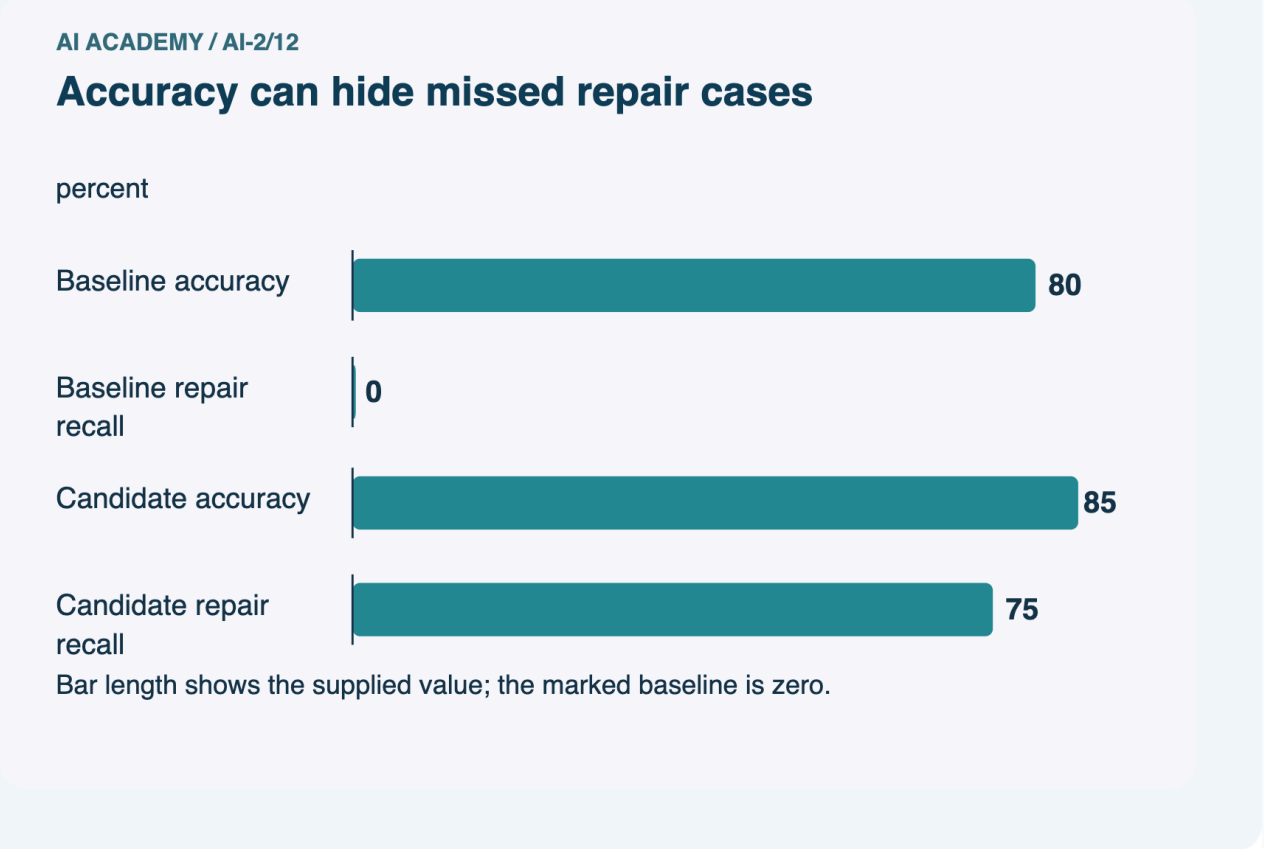


Follow the labelled picture with the matching explanation. Open the vector diagram

Reference population: 20 labelled cases, including four repair-needed books. → Baseline: 16/20=**80% accuracy**; catches 0/4 needed cases. → Candidate: (3+14)/20=85% **accuracy**; catches 3/4=75% needed cases. → Compare costs: Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Point to each numbered stage and explain which recorded **input** it uses.

03. Accuracy can hide missed repair cases

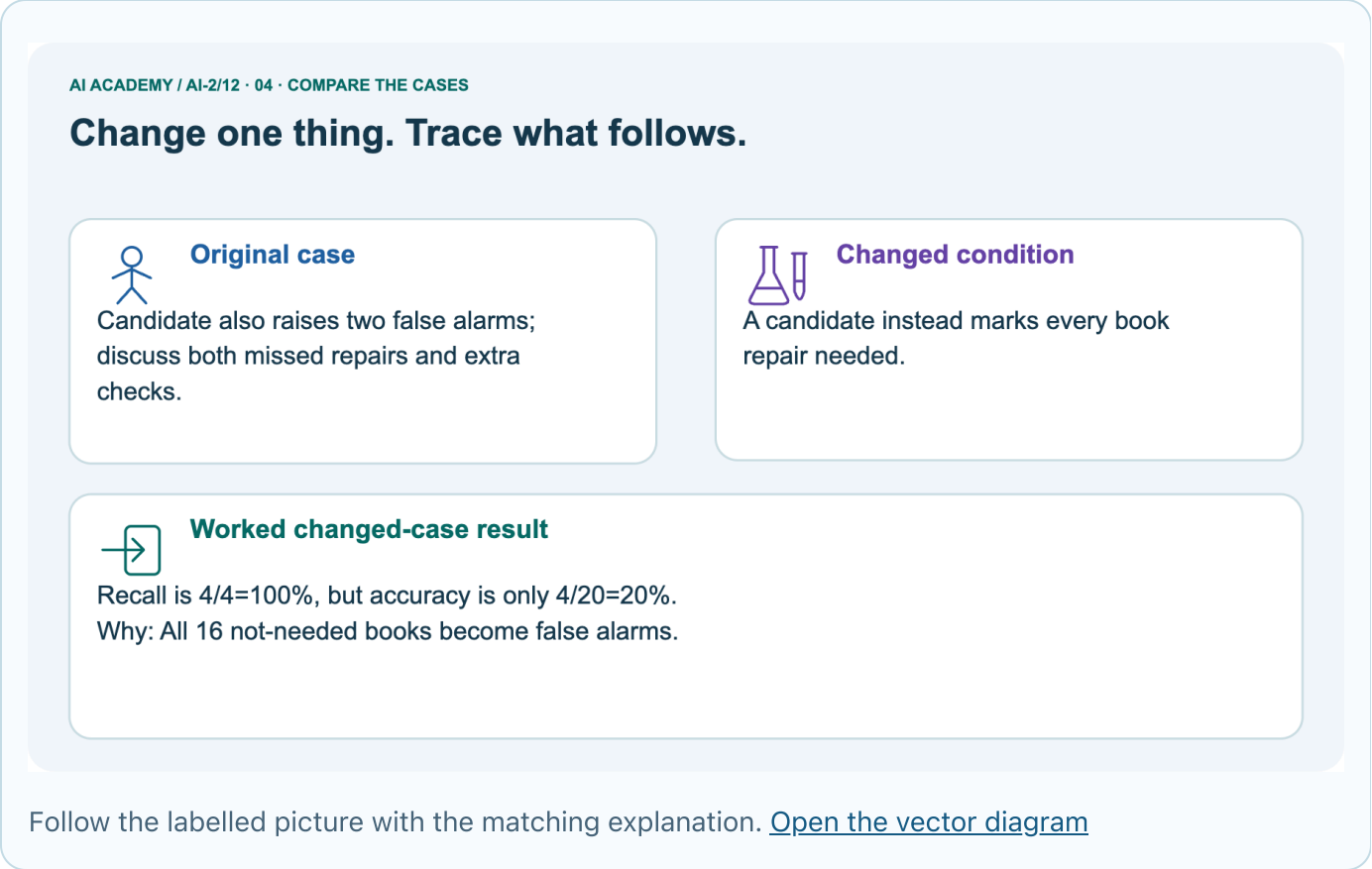


Follow the labelled picture with the matching explanation. Open the vector diagram

Accuracy can hide missed repair cases. Units: percent. Baseline **accuracy**: 80. Baseline repair recall: 0. Candidate **accuracy**: 85. Candidate repair recall: 75.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



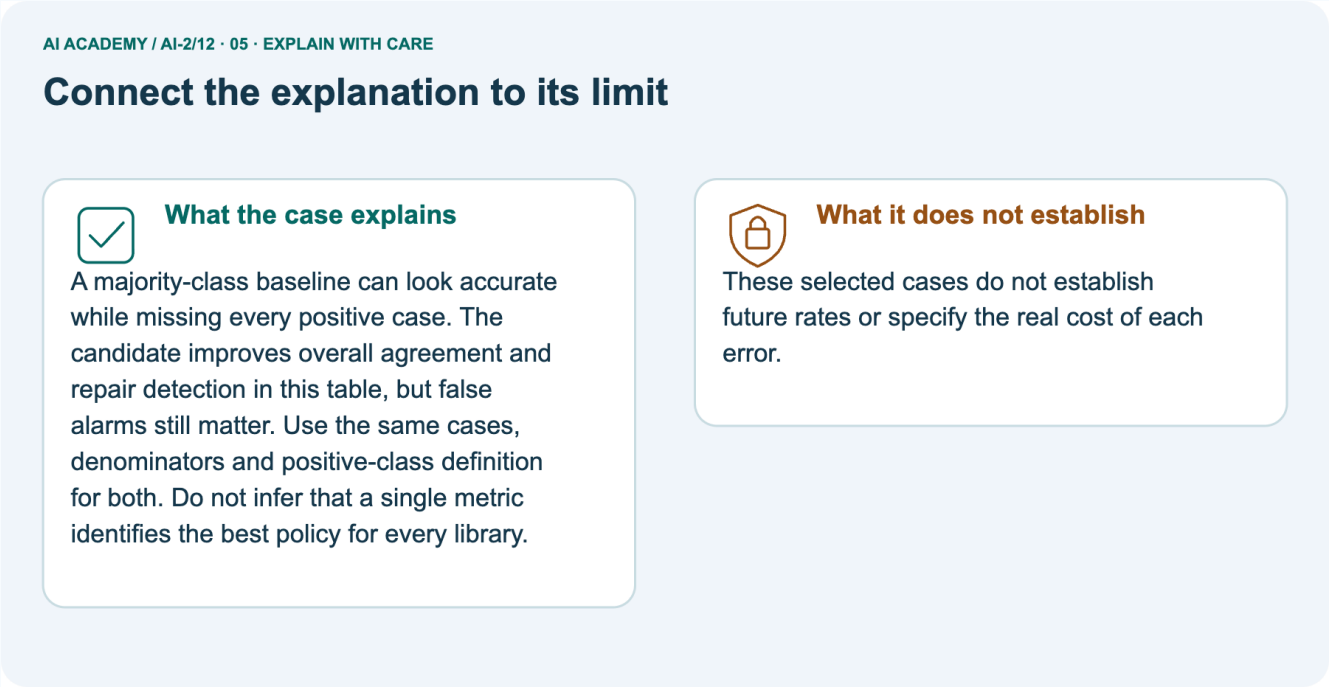
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Candidate also raises two false alarms; discuss both missed repairs and extra checks. Changed condition: A candidate instead marks every book repair needed.

Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. Reason: All 16 not-needed books become false alarms.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

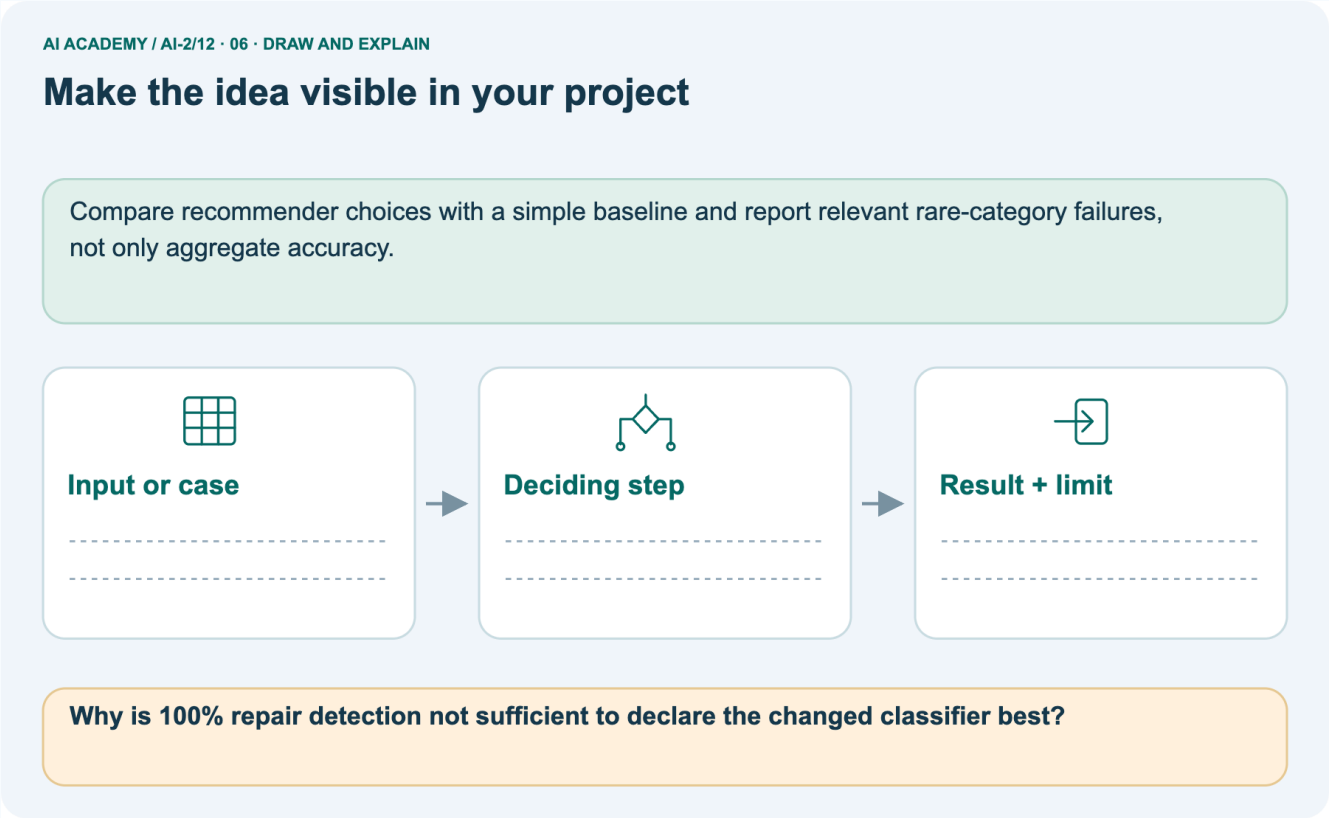
A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Limit: These selected cases do not establish future rates or specify the real cost of each error.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**. Draw your **input**, deciding step, result and limit.

Supported question: Why is 100% repair detection not sufficient to declare the changed classifier best?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Accuracy as a Fraction: the case

02. Follow the deciding steps

03. Accuracy can hide missed repair cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.